

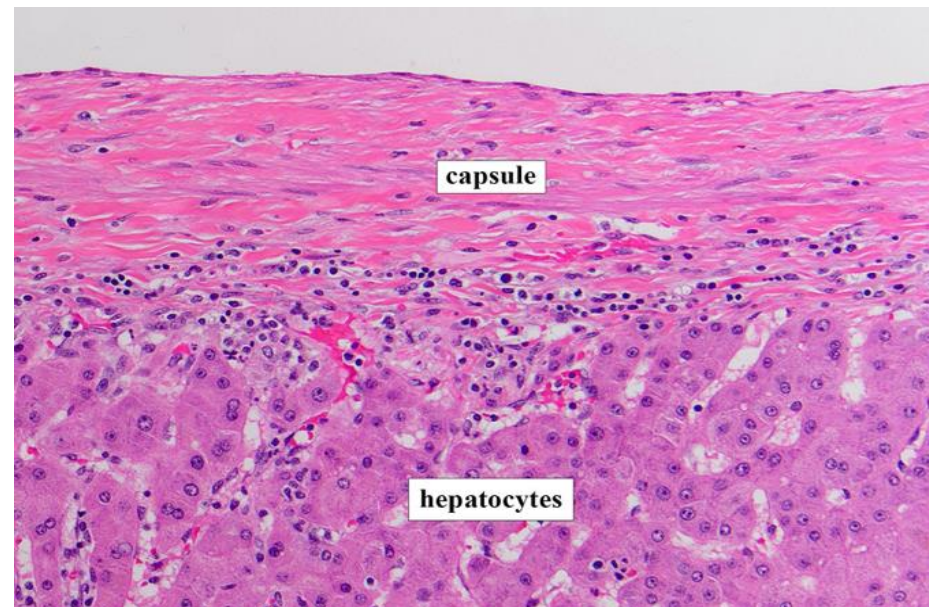
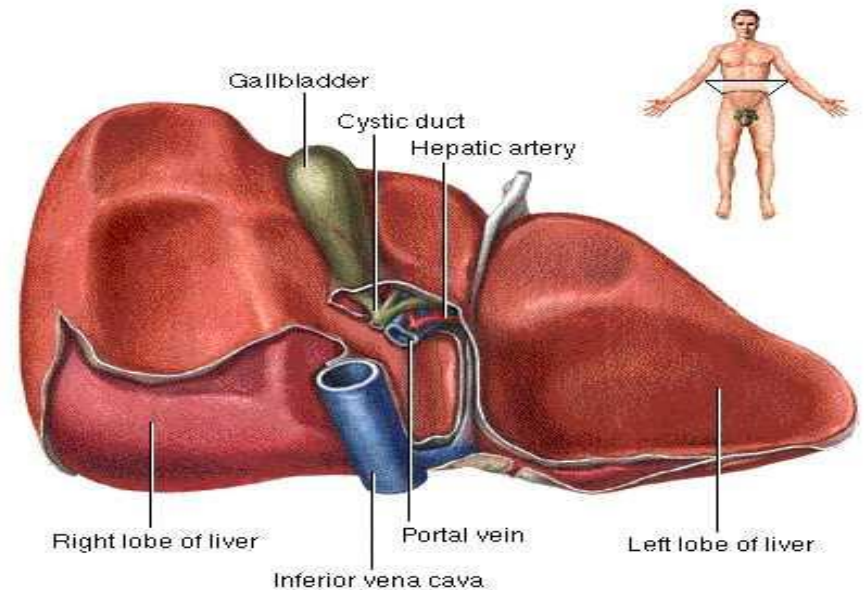
The Digestive System
Liver and Extra hepatic
biliary apparatus
SGL4

Dr. Hero K. Mustafa
Anatomy and Pathology department

Histology of the liver

Liver: largest gland in the body
Located in the abdominal cavity beneath the diaphragm, the liver has major left and right lobes with two smaller inferior lobes, most of which are covered by a thin capsule and mesothelium of the visceral peritoneum

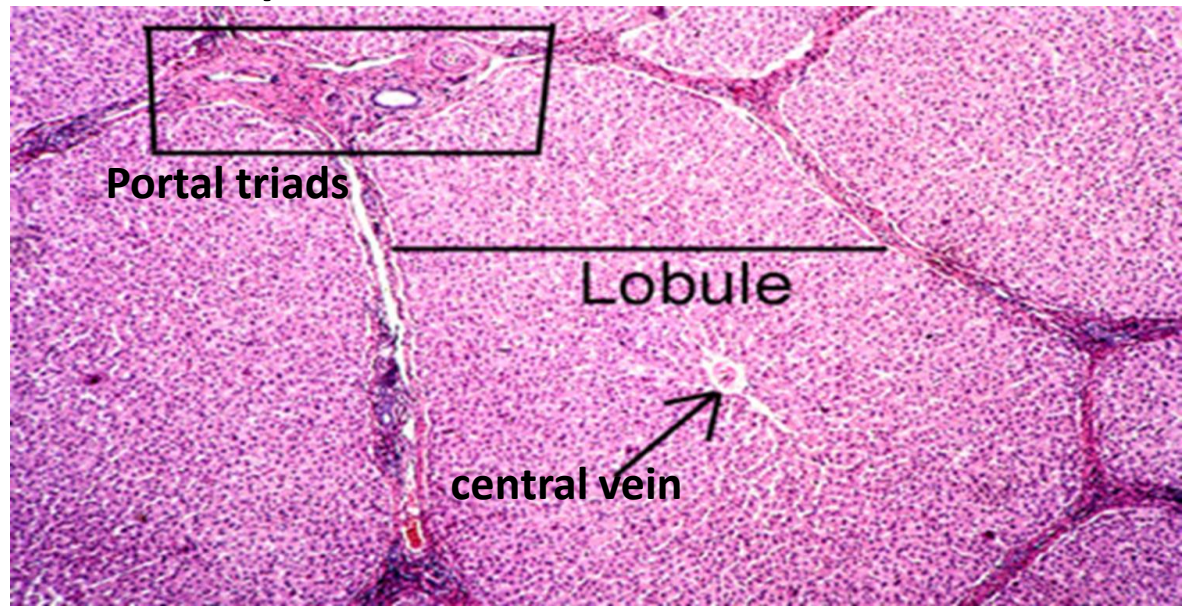
- Covered by c.t. capsule (**Glissons capsule**) that become thicker at hilum where the portal vein , hepatic artery enter & the right & left hepatic ducts and the lymphatic leave it

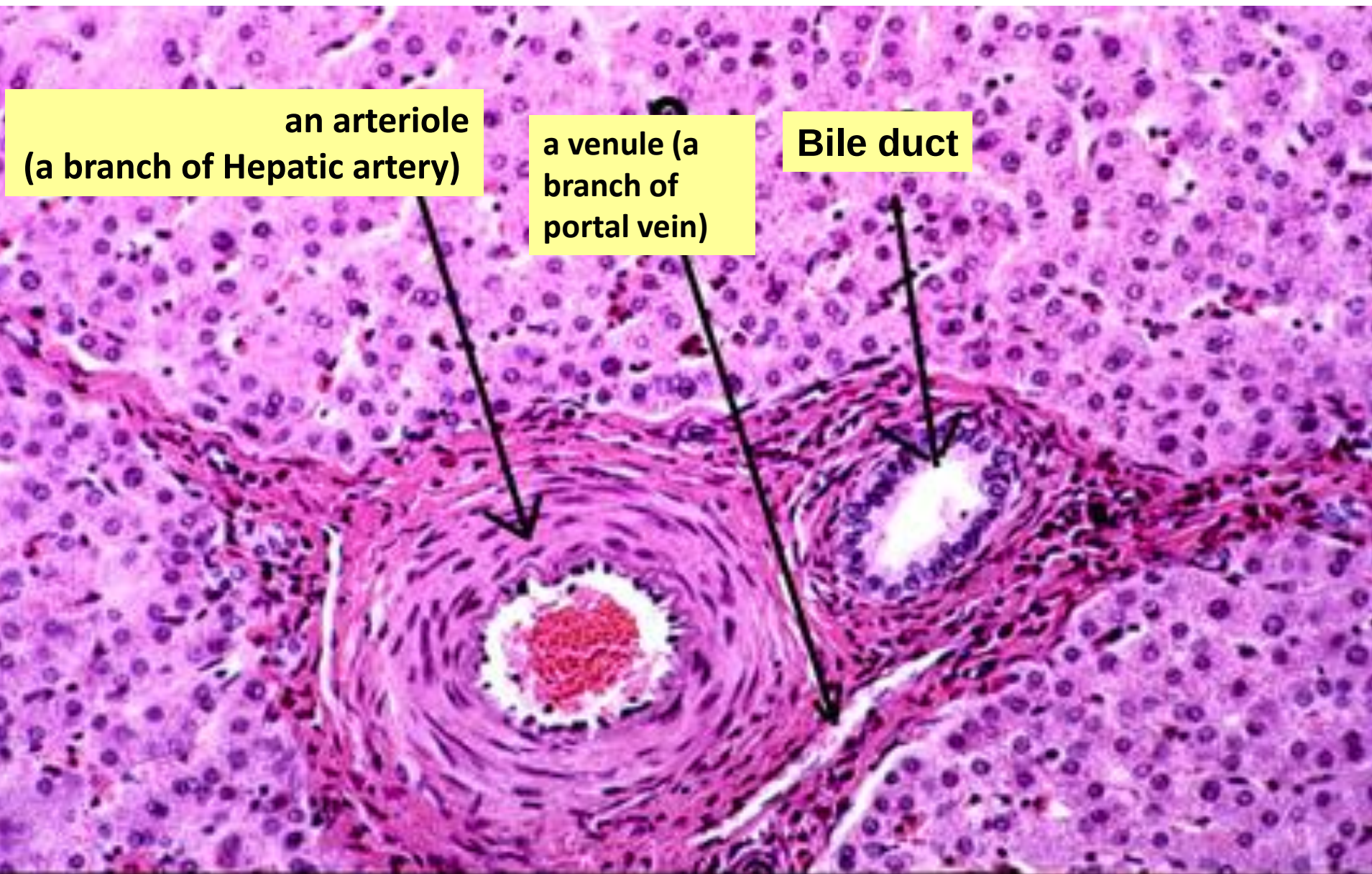


The parenchyma of the liver

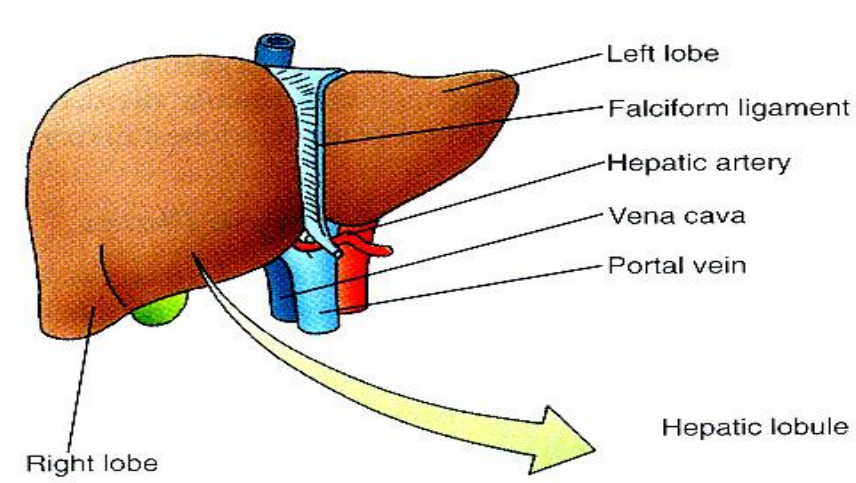
- consist of lobule , hexagonal-shaped
- Composed of hepatocyte (liver cell) radiating outward from a central vein
- Portal triads** are found at each of the six corners of each liver lobule, in human contain 3 to 6 portal triads per lobule. Portal triads consist of:
 - 1.-a duct (part of the bile duct system).
 - 2.-an arteriole(a branch of Hepatic artery)
 - 3.-a venule (a branch of portal vein)
 - 4.-Lymphatic vessels

These structures surrounded by c.t. together called **portal canal**

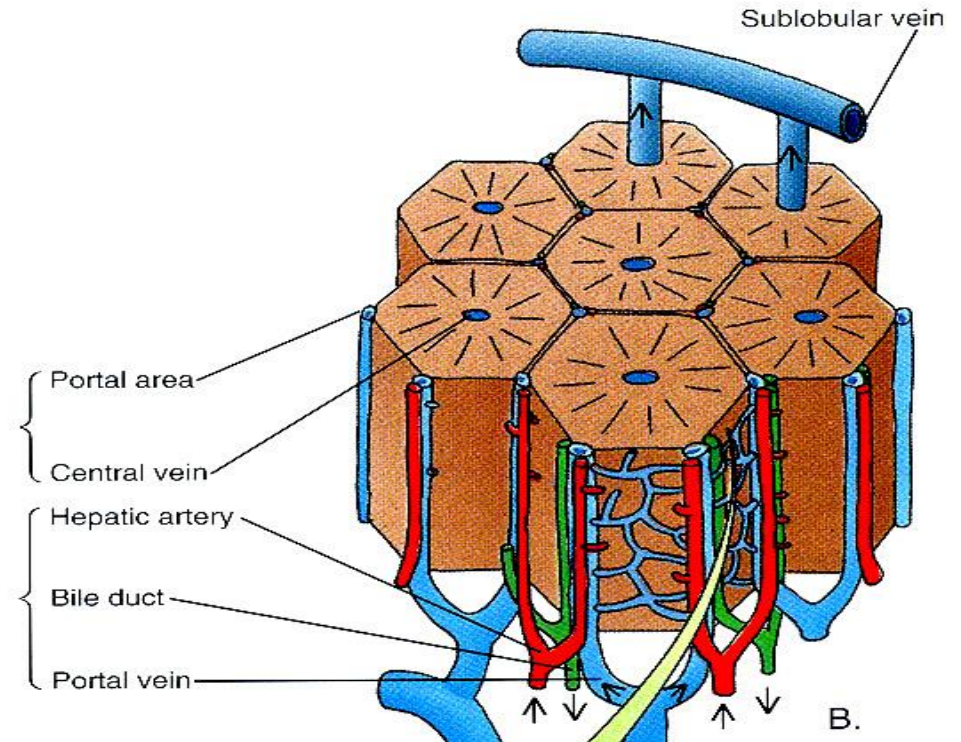




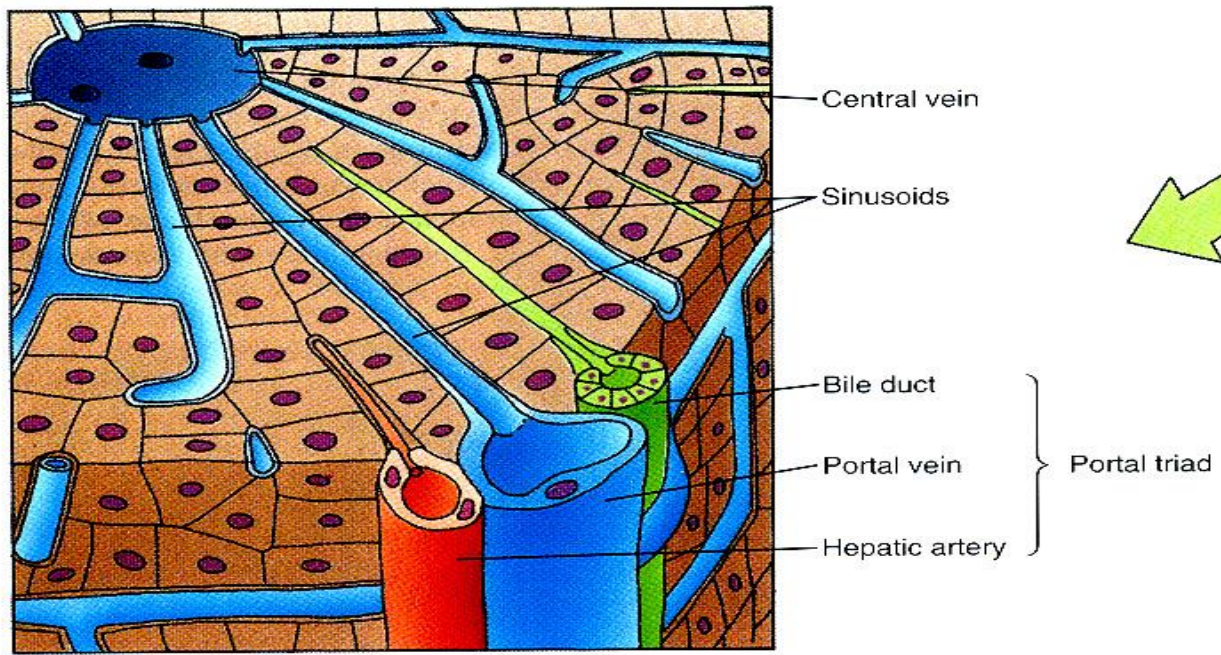
Portal triad



A.



B.



Cells in the liver

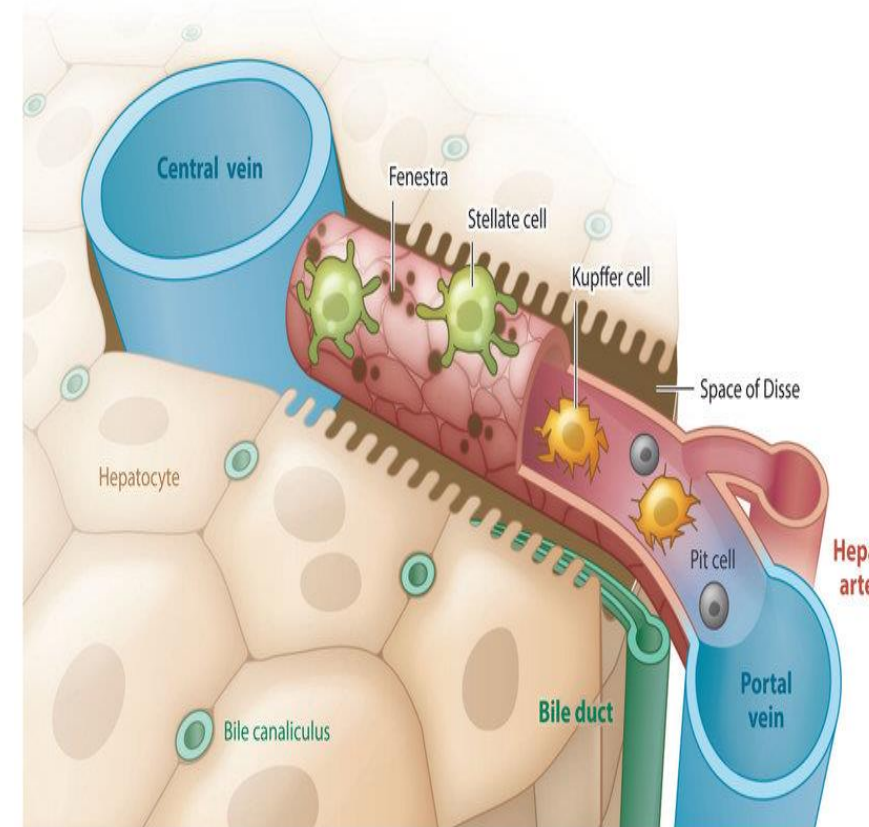
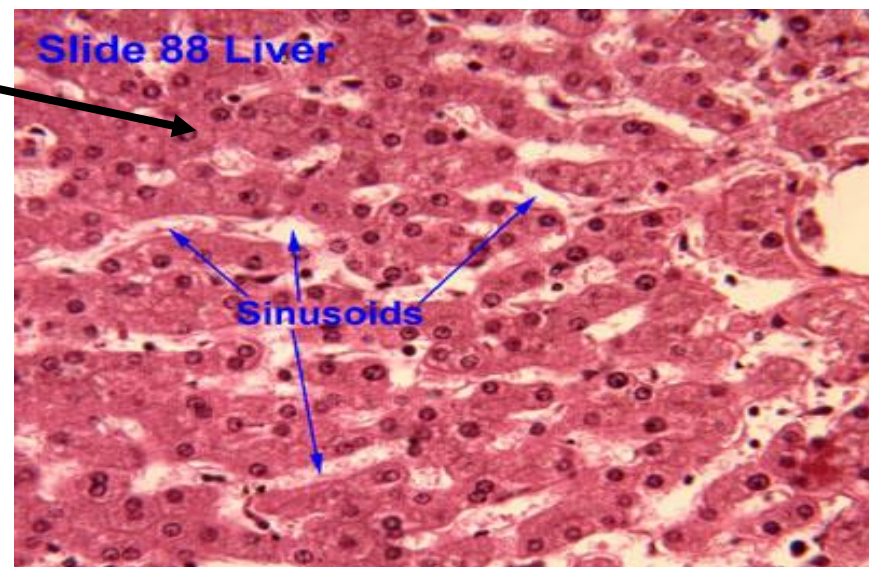
1-Hepatocytes:

These are the primary structure & functional unit of the liver.

are large, polyhedral cells with 6 or more surfaces, eosinophilic cytoplasm rich in mitochondria and organelles. Cells are arranged in the form of plates anastomose with one another to form a network. Spaces within the network are occupied by sinusoids. • In this way each liver cell has a sinusoid on two sides.

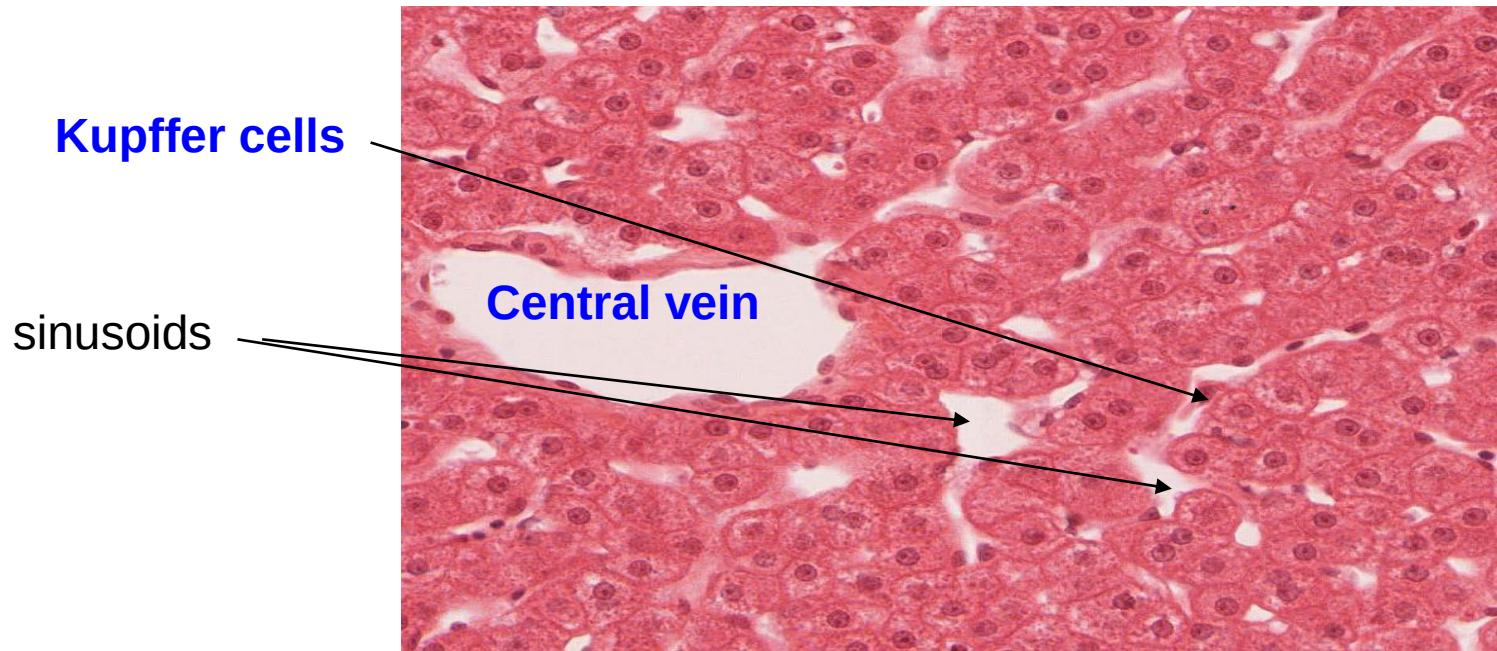
The sinusoids are irregular dilated vessels composed of discontinuous layer of fenestrated endothelial cells and basement membrane.

• The surface of the liver cell is separated from the endothelial lining of the sinusoid by a narrow perisinusoidal **space of Disse**. • Microvilli, present on the liver cells extend into this space.



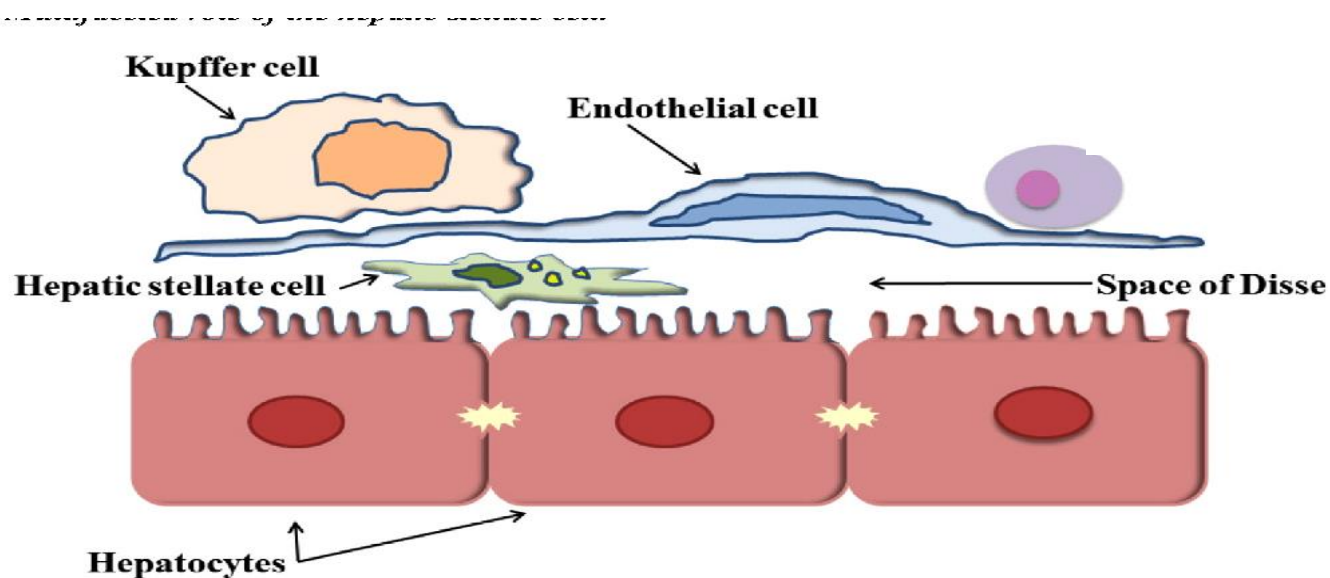
2-Kupffer cells : Are monocyte derived cells

- Found on the luminal surface of the endothelial cells
- Contain ovoid nucleus , & organelles
- Usually contain engulfing materials
- **Main function:**
 1. **Phagocytose aged erythrocytes, freeing heme and iron for reuse or storage in ferritin complexes.**
 2. **antigen-presenting cells and remove any bacteria or debris in the portal blood.**
 3. **produce %5 of proteins**



- **3-(Ito-cells) fat-storing cells:**

- Ito cells, also known as fat-storing cells or hepatic stellate cells, are located in the space of Disse in the liver
- these cells are characterized by the presence of lipid droplets in their cytoplasm.
- Function:
 1. Their primary function is to **store vitamin A in these droplets**
 2. these cells are a major **source of collagen and other extracellular matrix** components, which they secrete, particularly during liver injury.
 3. regulating blood flow in the sinusoids **by long, contractile processes**
 4. they are involved in **producing various cytokines** that help regulate **Kupffer cell** activity.



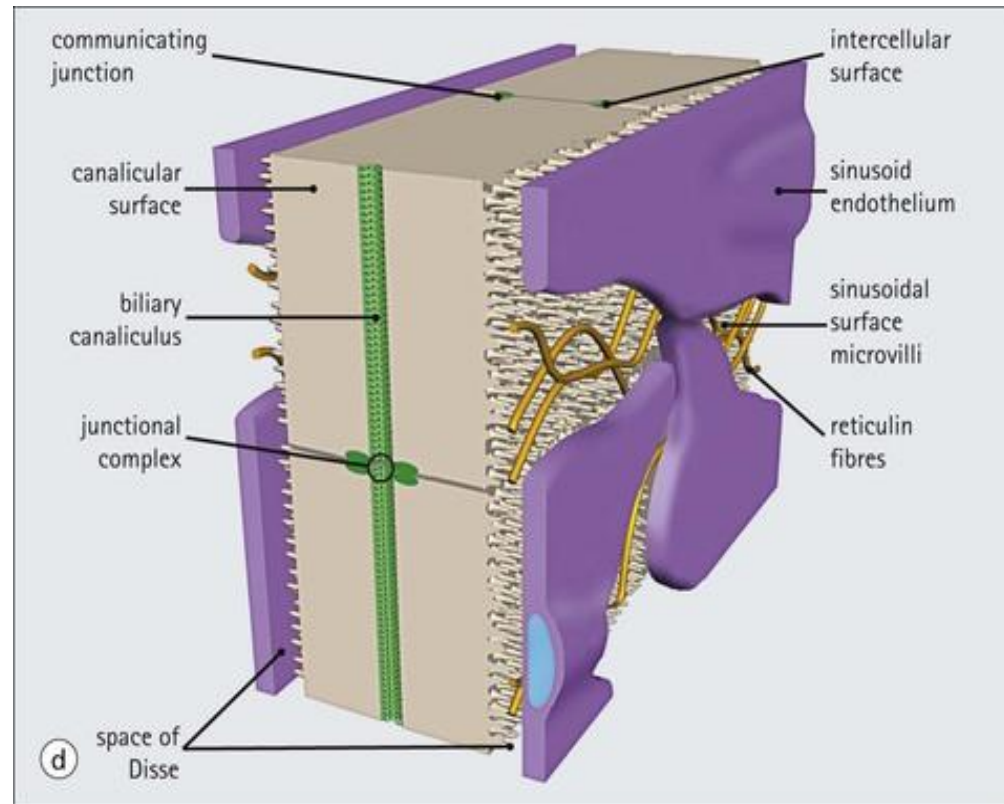
1 - Hepatic stellate cell location in liver parenchyma

The surface of hepatocyte has three types of specializations

1-Sinusoidal surface :which faces to the blood sinusoids have microvilli that project into the space of Disse... responsible for exchanging materials with the blood

2-Canalicular surface:

this area of cell membrane have longitudinal depression on neighboring hepatocytes, to form the wall of a bile canaliculus. Irregular microvilli project into the canaliculus.it's the primary site for bile secretion from the hepatocyte into the biliary system.

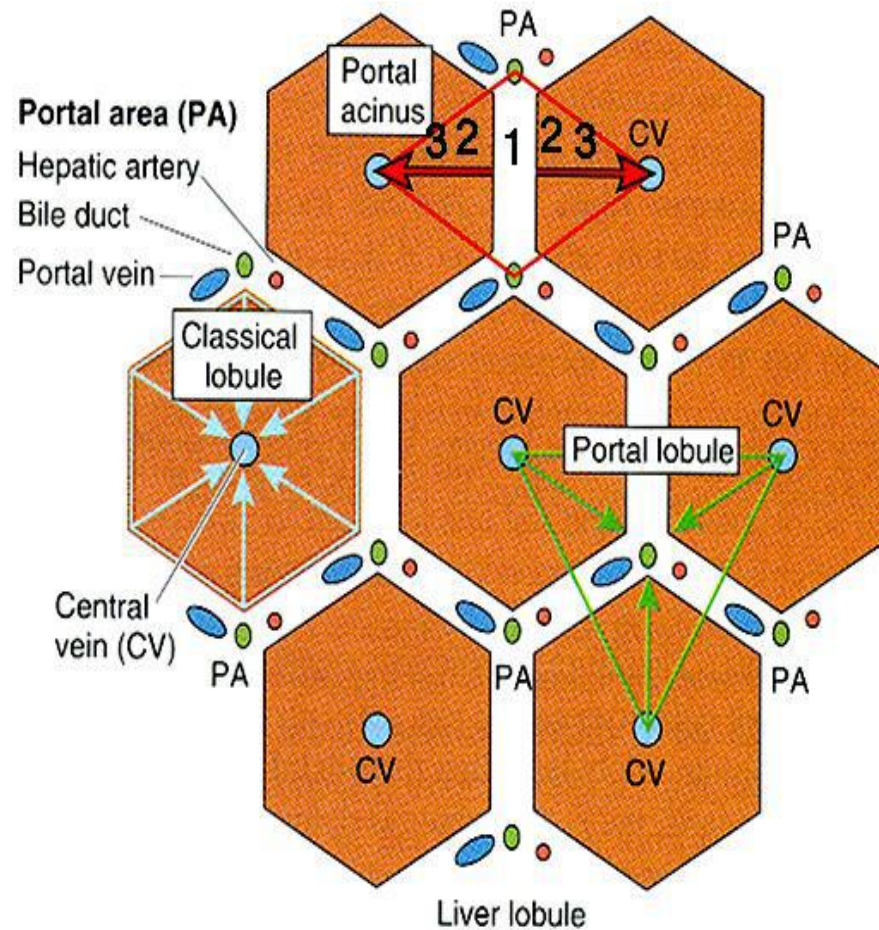


3-Intercellular surface:

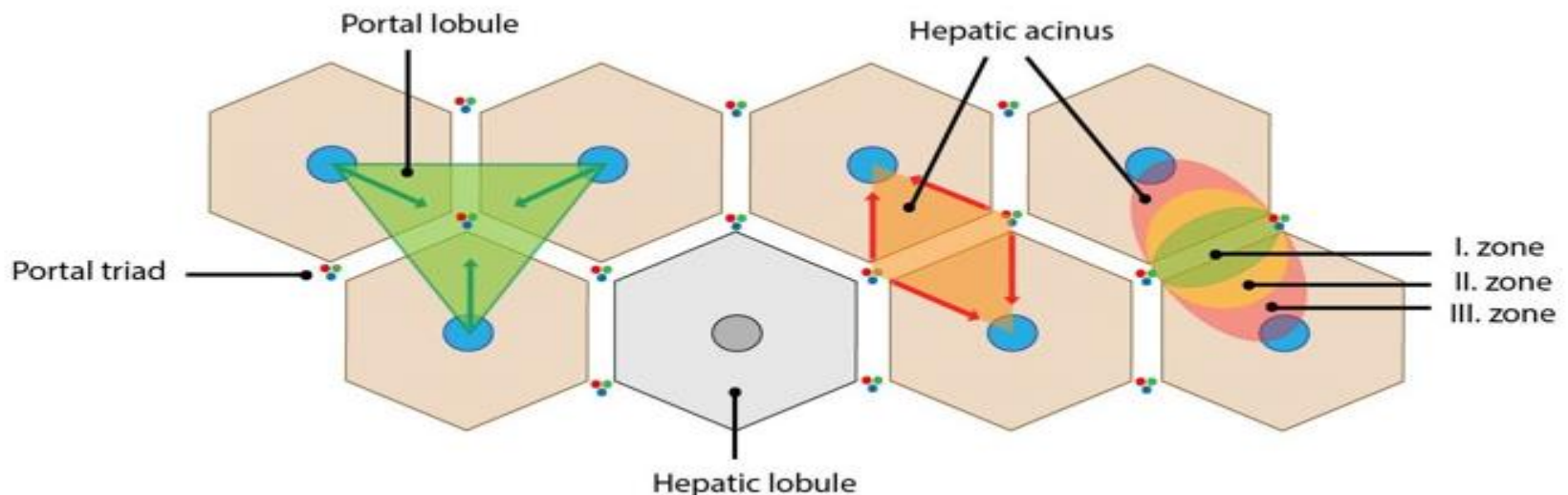
which is the area of contact between adjacent hepatocytes, where various cell junctions are located, allow exchanges between the cells.

subdivision of the liver

- **1-Classic Lobule** hexagonal - shaped lobule, portal triad at each corner & a center vein in the center of each lobule.
- This model is based on the **direction of blood flow.**
- **2-The portal lobules:**
- This model is based on the **direction of bile flow which is opposite to that of blood.**
- The liver parenchyma is divided into interlocking triangle ,each with portal triad at its center & a central vein at each of its three corners.



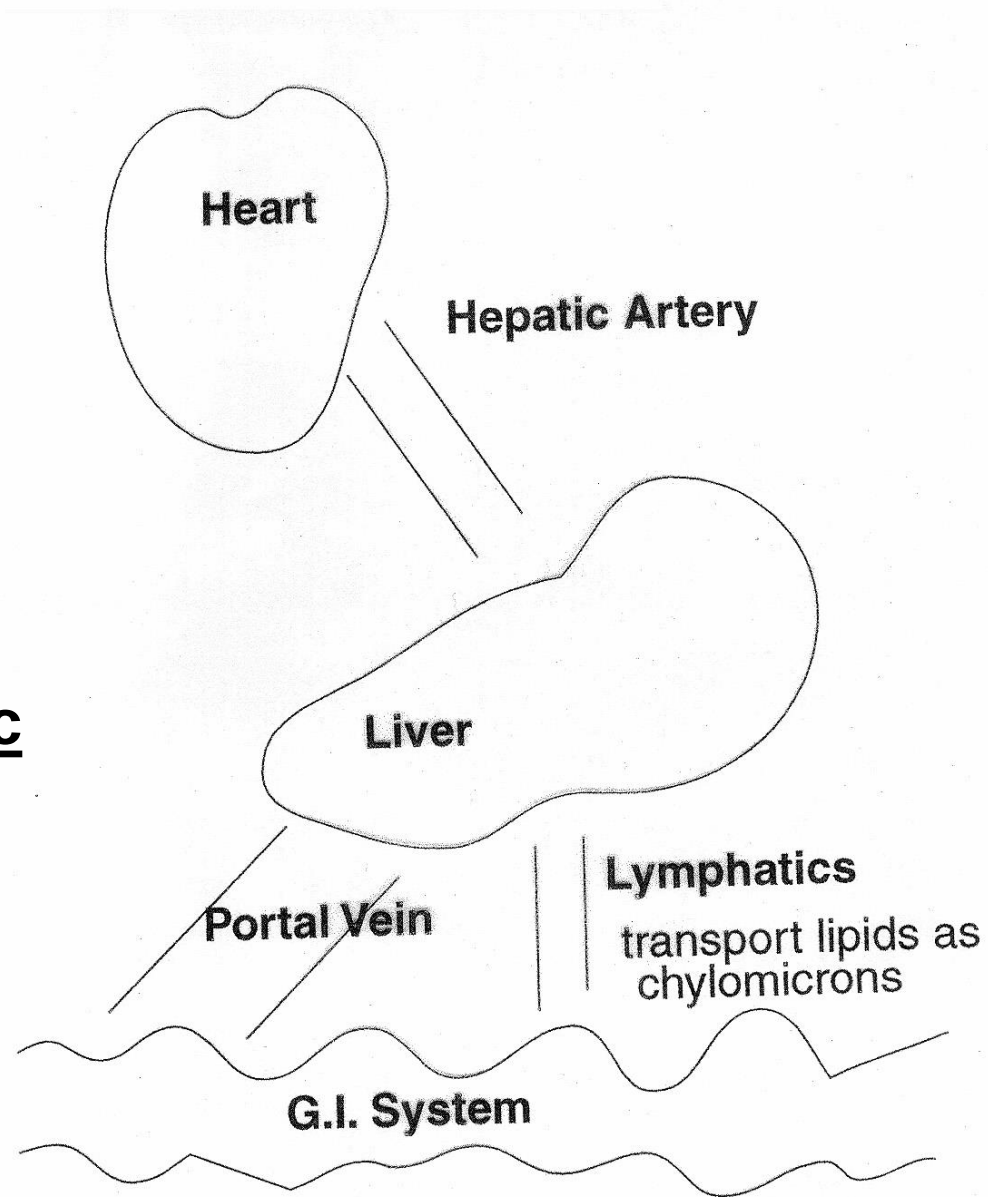
- **3-The hepatic acinus**
- Is functional lobules based on changes in oxygen, nutrient & toxin content as blood flowing through the sinusoids.
- Each appear as diamond shape contain two central veins & two portal triads, that define its four corners.
- The diamond is divided into two triangles by a line connecting the portal triads. Each triangle can be divided into three zones, according to there distance from the terminal distributing vessels.
- Zone I, for example is closer to these vessels where as Zone III is closer to the central vein. Blood in Zone I sinusoids has higher oxygen, nutrients & toxin concentration than in the other zones.



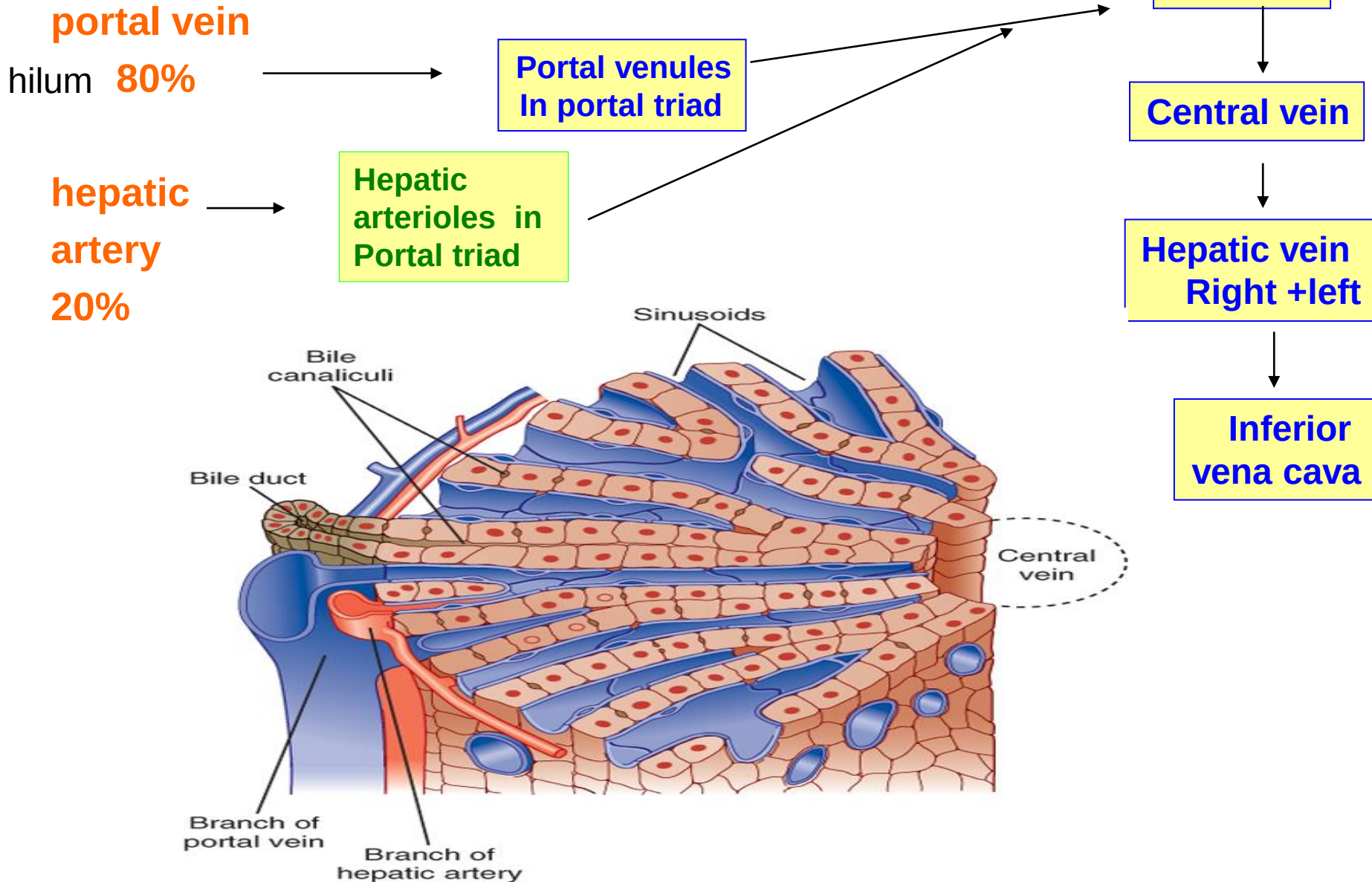
Blood circulation in the liver

80% of liver's blood is supplied by portal vein conveys blood from the pancreas, spleen and intestine, contain oxygenated poor with nutrient rich blood

20% is supplied by hepatic artery, is higher oxygenated blood



Blood circulation in the liver



Bile duct system:

- The exocrine secretion of the hepatocytes is called bile. Hepatocytes promote the uptake, transformation and excretion of blood components into the bile canaliculi
- **Components of bile:**
- Water, electrolytes, bile acids, phospholipids, cholesterol and bilirubin
- About 90% of these substances are derived by absorption from the intestine
- About 10% of bile acid are synthesized in the smooth endoplasmic reticulum of hepatocytes
- Bile acids have an important function in emulsifying the lipids in the digestive tract prompting easier digestion by lipase and absorption

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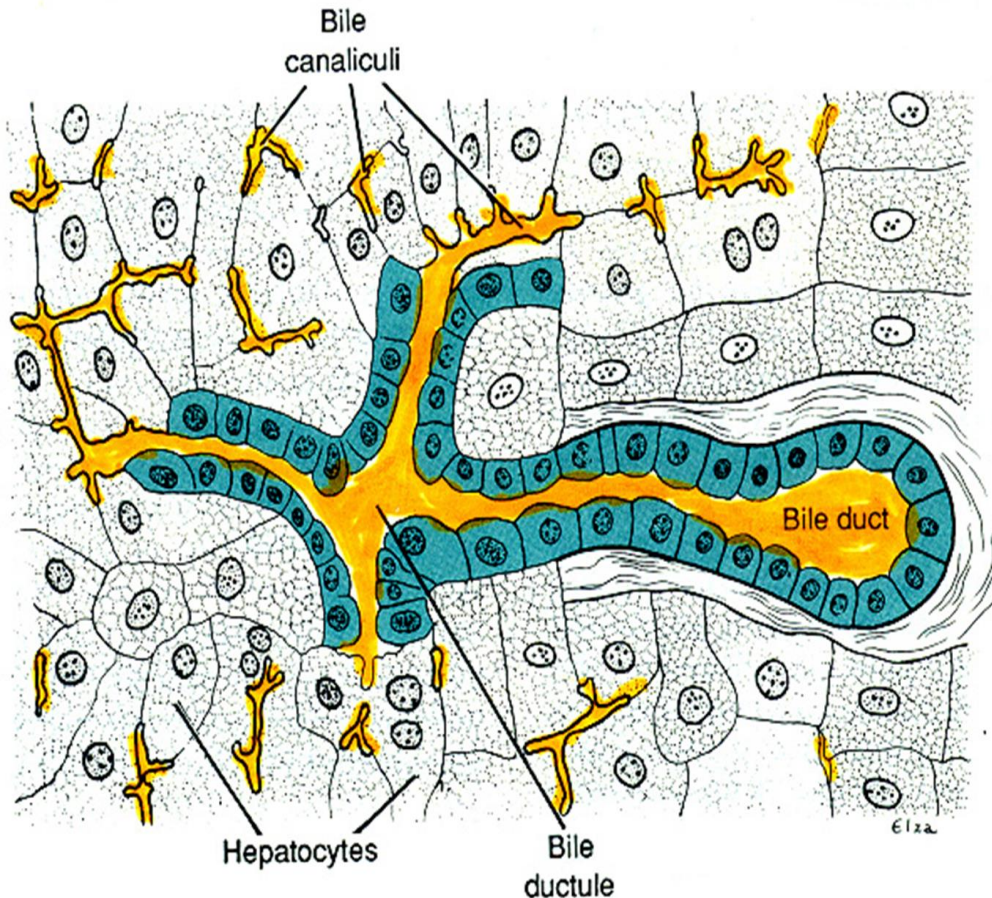
Bile duct system

Bile canaliculi
they are limited
only by plasma
membrane of
two hepatocytes

**Bile
ductules
or Herings
canals
(cuboidal)**

**bile ducts
(cuboidal to
columnar)
in portal
triad**

**Right & left
hepatic ducts**
Leave liver



**Cystic duct
From gall bla.**

**Common hepatic
Ducts in hilum**

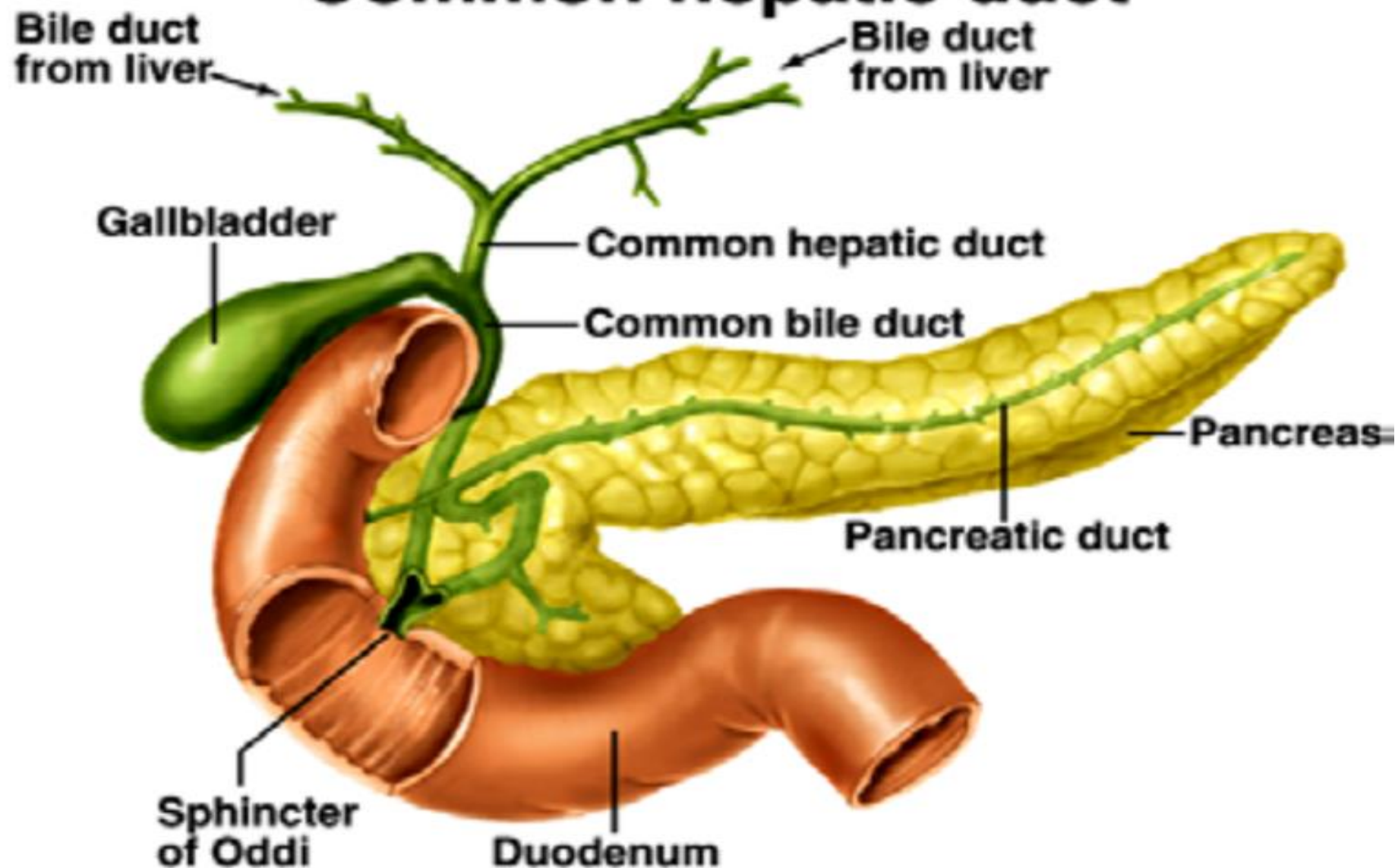
**Pancreatic
duct**

**Common
bile
duct**

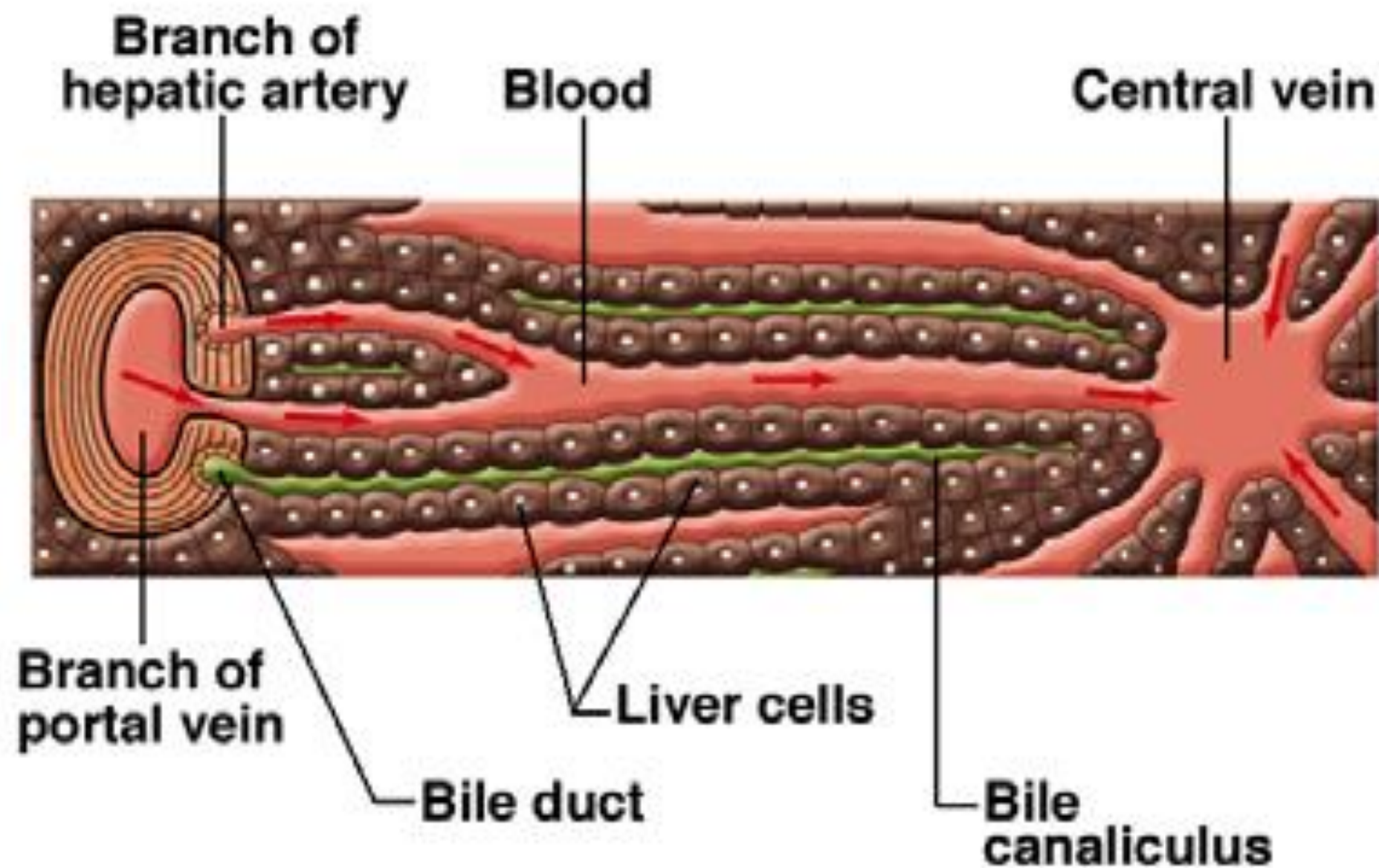
duodenum

Ampulla of vater

Common hepatic duct



Bile canaliculi and ducts

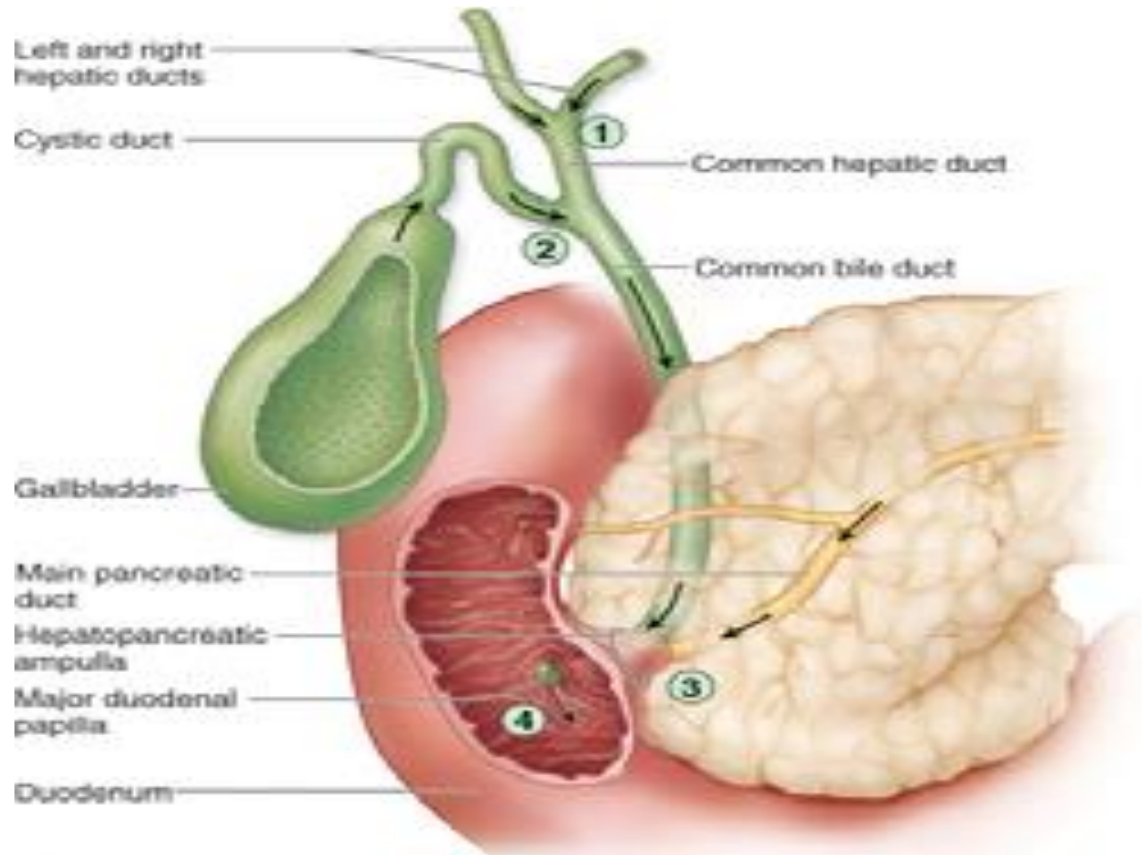


Extrahepatic biliary apparatus

Includes:

**1-extrahepatic
bile ducts**

2- Gall bladder



① Left and right hepatic ducts merge to form a common hepatic duct.

② Common hepatic and cystic ducts merge to form a common bile duct.

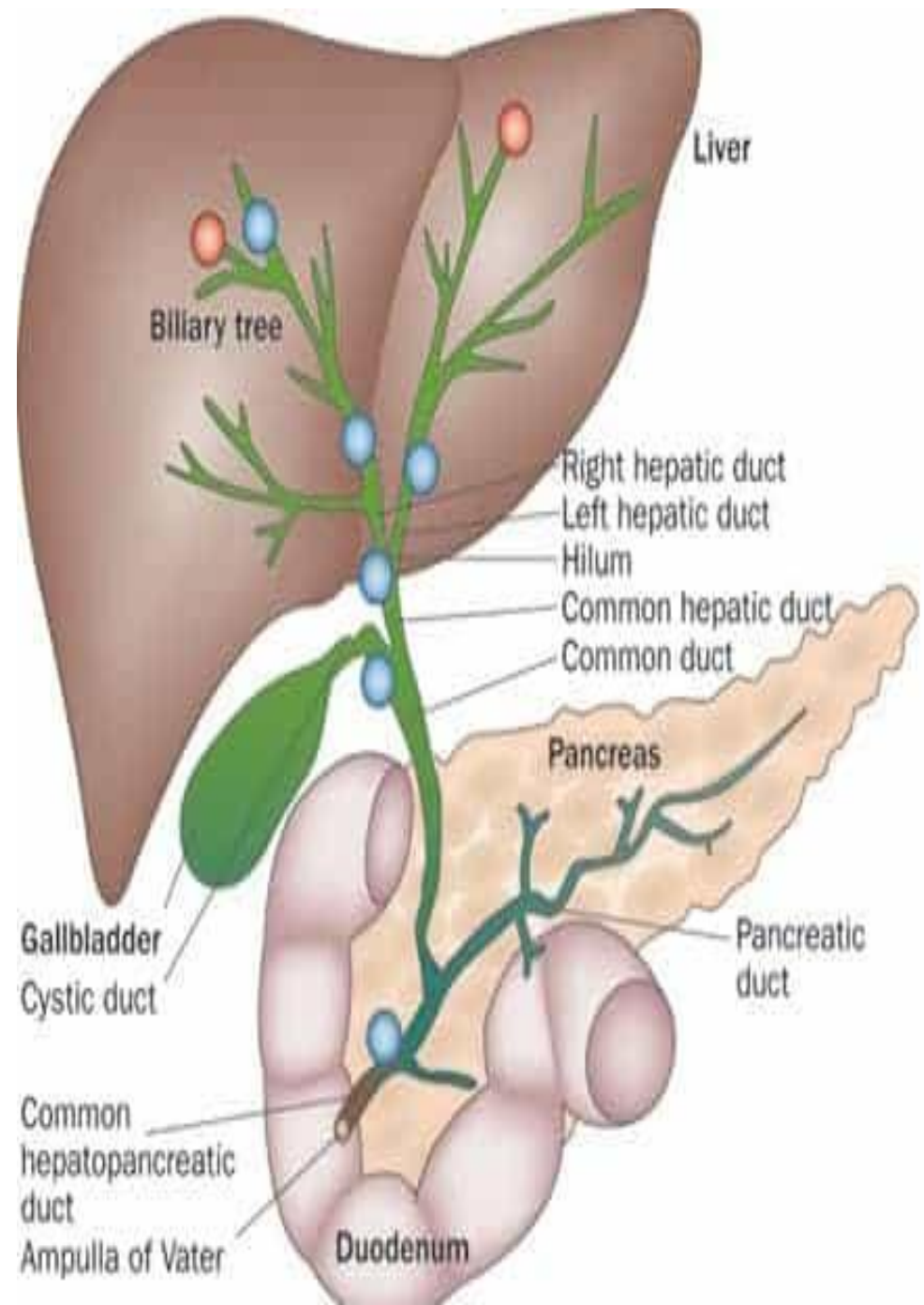
③ Pancreatic duct merges with common bile duct at the hepatopancreatic ampulla.

④ Bile and pancreatic juices enter duodenum at the major duodenal papilla.

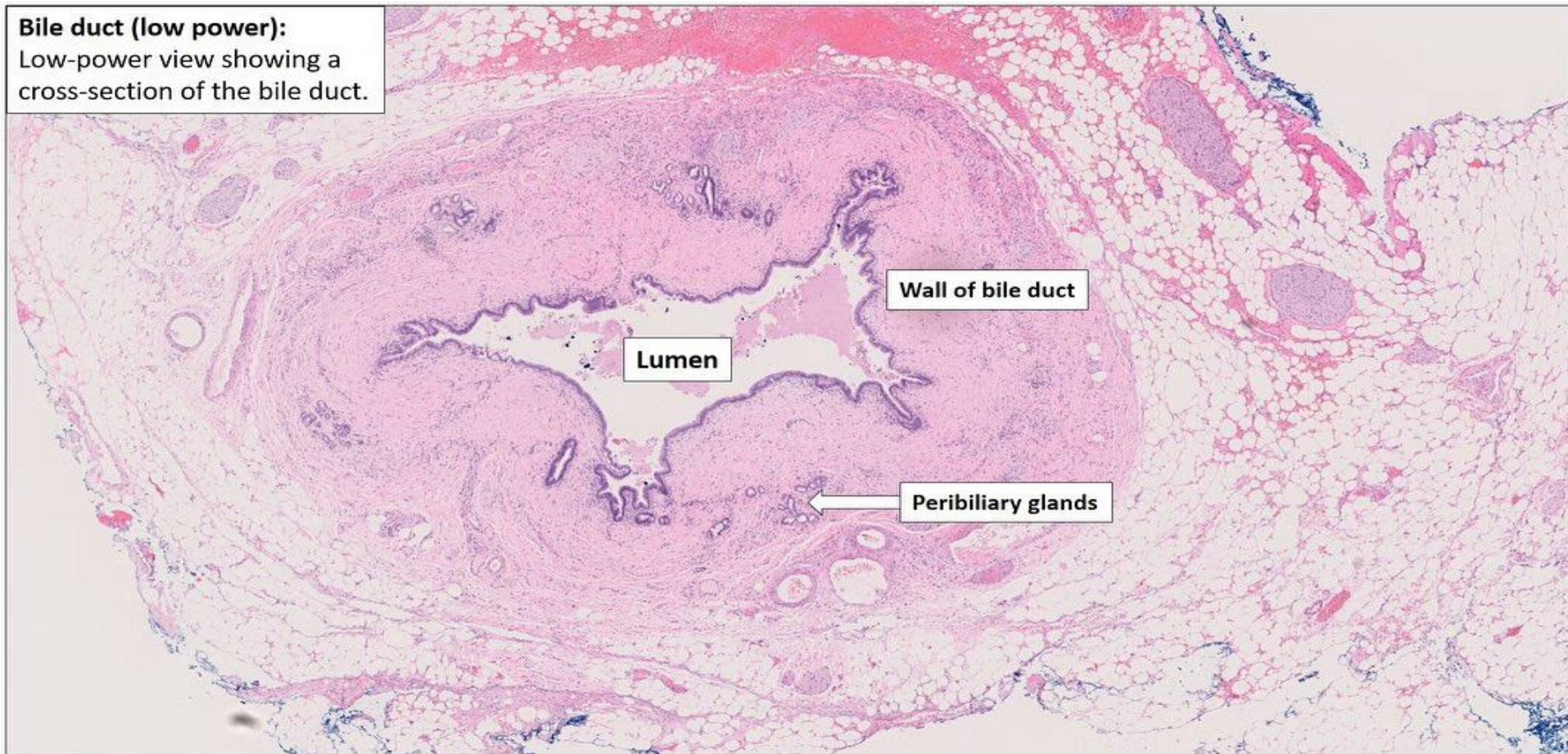
1-Extra hepatic bile ducts

- Right, left & common hepatic ducts
- Cystic duct
- Bile duct
- Lower end bile duct joined with main pancreatic duct forming a common hepato-pancreatic

duct or ampulla (ampulla of Vater) opens into the duodenum, regulated by the sphincter of Oddi.

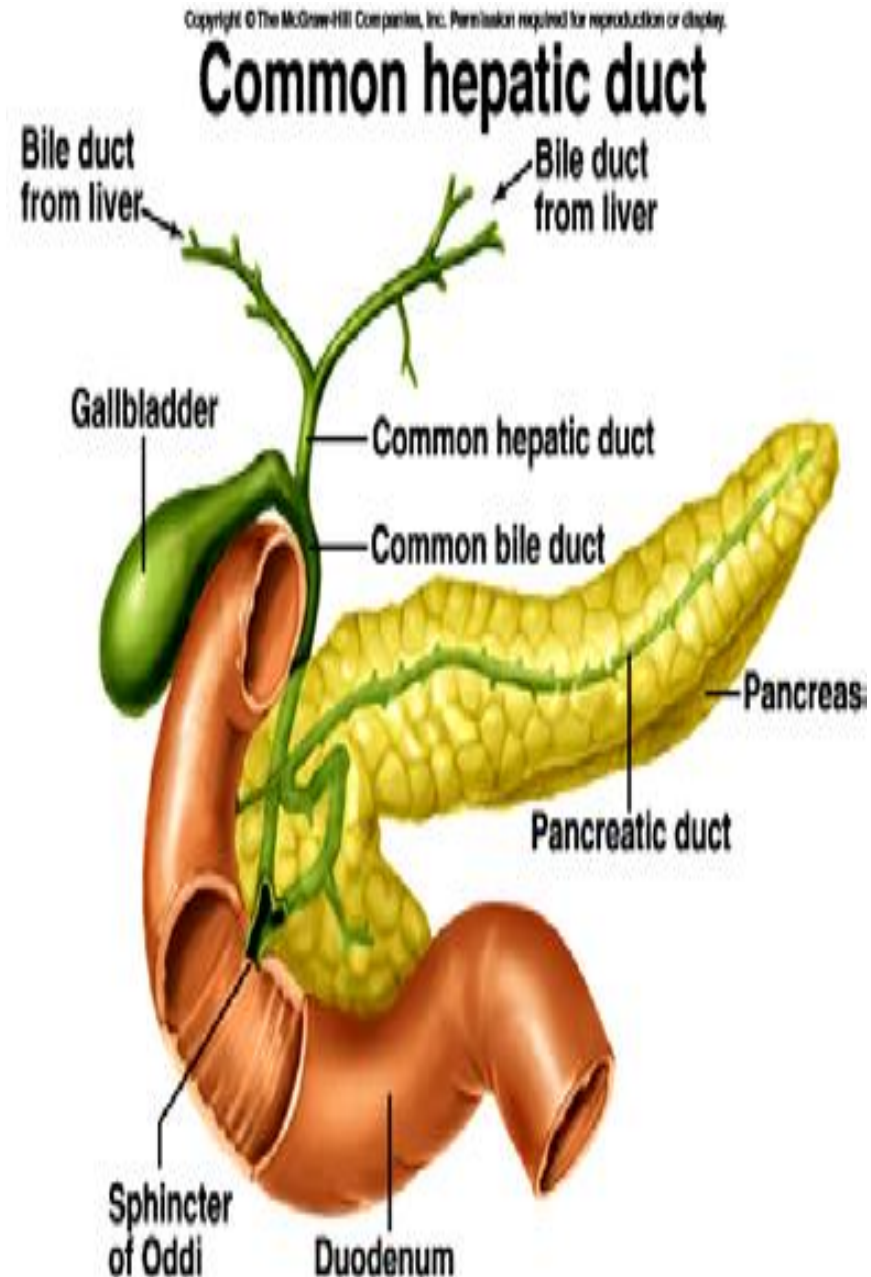


The hepatic, cystic, and common bile ducts are lined with a mucous membrane having a simple columnar epithelium. and surrounded by a thin muscularis. This muscle layer becomes thicker near the duodenum and finally, in the duodenal papilla, forms a sphincter that regulates bile flow(sphincter of Oddi).



2-Gallbladder

- pear-shaped sac under the liver
- Store & concentrated the bile
- Releases bile via the bile duct
- Secretion of Bile is regulated by the hormone Cholecystikin (CCK)
- Their wall consist of :
 1. Mucosa
 2. Muscularis
 3. Serosa& adventitia



Histology of Gallbladder

Mucosa: highly folded, simple columnar cells with microvilli (no goblet cells), epithelia supported by lamina propria

Submucosa: similar to GI - blood vessels, lymphatics & nerves

Muscularis: two or three irregular layers smooth muscles

Serosa / adventitia: mesothelium / adventitia where gallbladder contacts liver



Thanks